Real-Time Closed-Loop Brain-Body Simulation Using a GPU-Based Multi-Region Spiking Neural Network and a Mouse Musculoskeletal Model

GPUベース多領域SNNとマウス筋骨格モデルを使ったリアルタイム閉ループ脳身体シミュレーション

Masachika Taniguchi, Michiru Inoue, Tadashi Yamazaki

Graduate School of Informatics and Engineering, The University of Electro-Communications

GPU-based SNN ran the brain–body loop faster than real time overall.

10.05 s simulated in 9.38 s; 50-ms blocks averaged 46.6 ms.

INTRO

- Motor control depends on closed-loop brain–body interactions with sensory feedback.
- We built a real-time closed-loop simulation by coupling a GPU-based multi-region SNN to a mouse forearm model

METHODS

- A GPU SNN simulated M1/S1, BG, TH, cerebellum, and CPG.
- M1_L5B_PT activity drove antagonist extensor–flexor muscles.
- Body motion was fed back to S1, closing the loop.

RESULTS

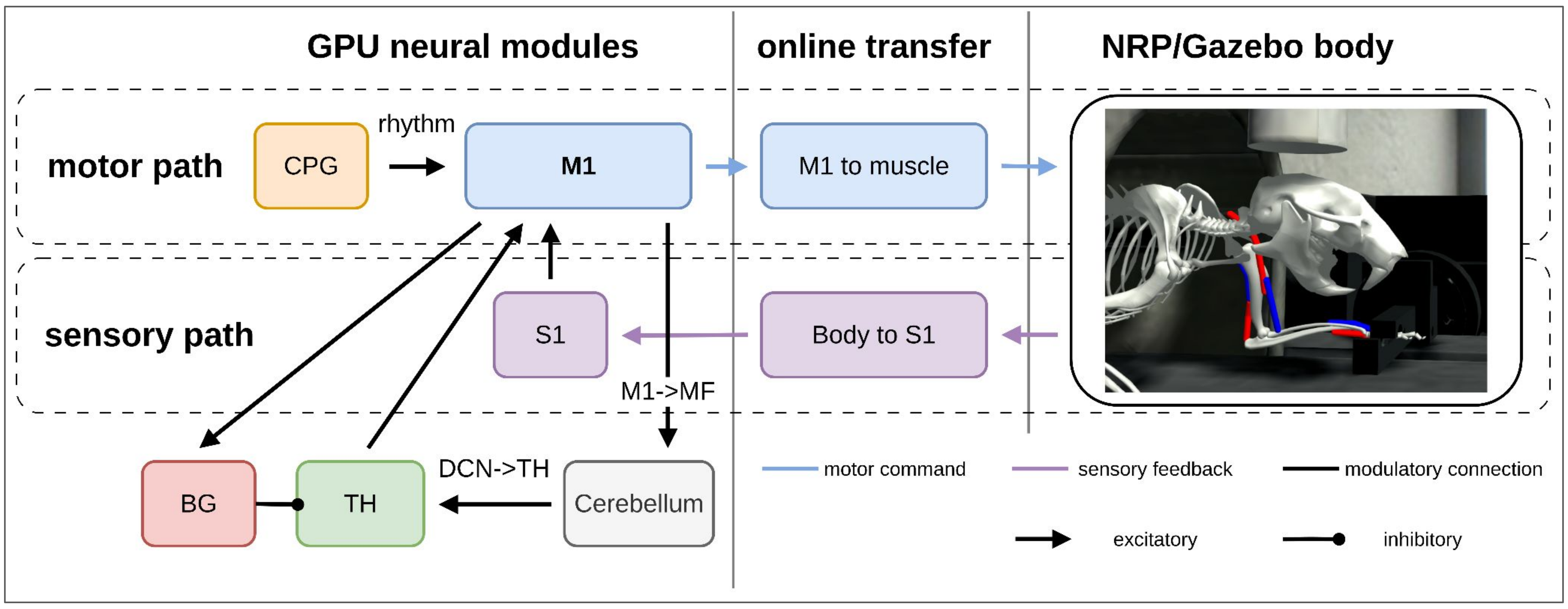
- M1 output generated reciprocal muscle activation and periodic forearm/lever motion.
- S1 activity was modulated by body motion.
- The loop ran faster than real time overall: 10.05 s in 9.38 s.

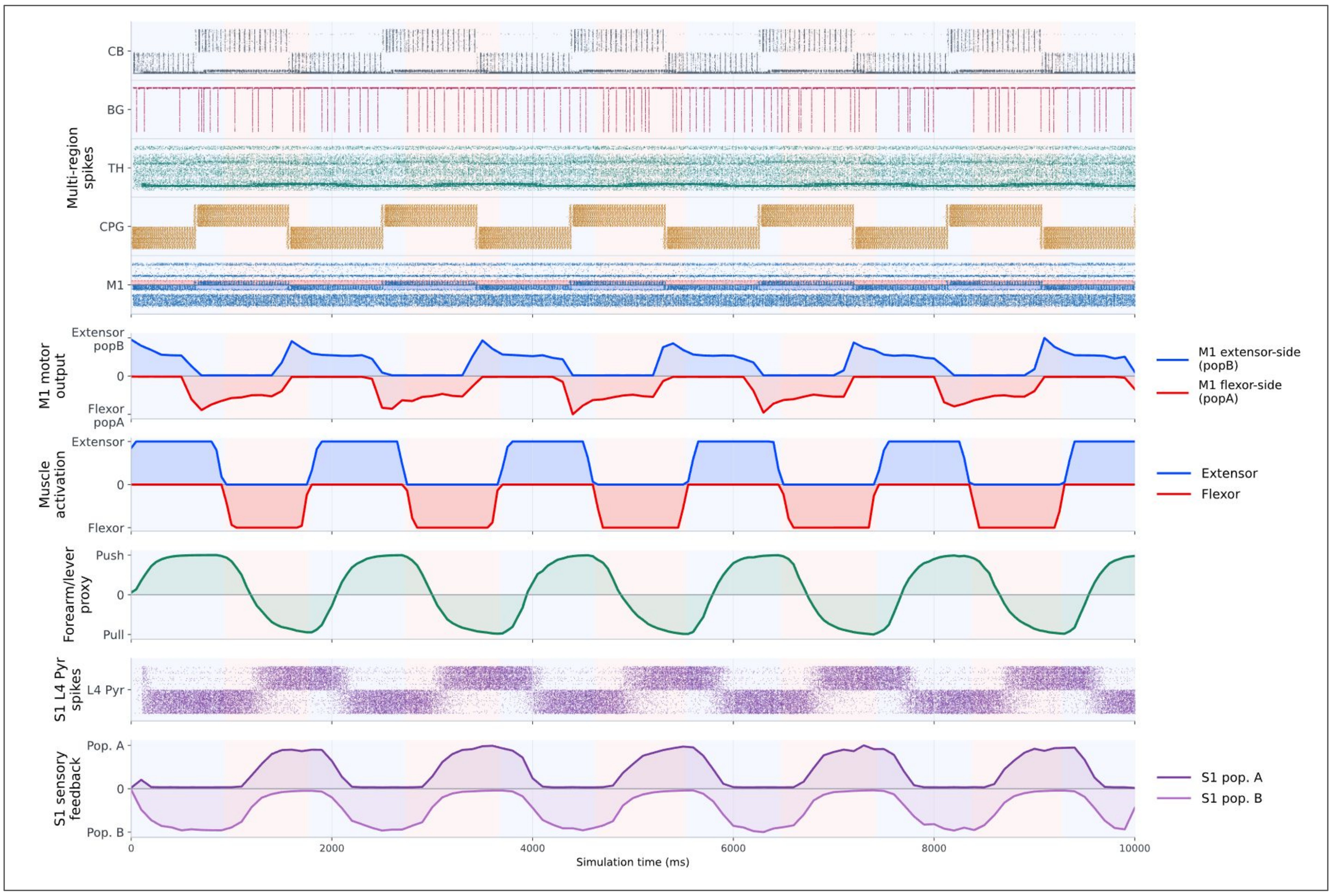
CONCLUSION

- GPU execution enabled an overall faster-than-real-time closed-loop brain–body simulation.
- The framework supports testing sensory feedback, cerebellar interactions, and cortico–basal ganglia loops in movement generation.

REFERENCES

1. Kuniyoshi et al., Front. Neurorobot., 2023.
2. Falotico et al., Front. Neurorobot., 2017.
3. Kuriyama et al., Front. Cell. Neurosci., 2021.





Simulation / wall-clock
1.07x
10.05 s simulated / 9.38 s wall

Loop latency
46.6 ms
mean; target = 50 ms per block

Blocks within target
87%
174/201 blocks <= 50 ms; max 58.3 ms